Final Project Kickoff: gpt-oss

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MOREH

Project Overview

- **Goal:** Implement high-performance inference for gpt-oss models (20B & 120B parameters)
- What is llama2.c?
  - A pure C inference engine for LLaMA 2, running on CPU
  - We extended to support gpt-oss
  - https://github.com/karpathy/llama2.c
- What is gpt-oss?
  - OpenAI's open-source large language model
  - Parameter sizes: 20B, 120B
    - Interactive (chat)
    - Non-interactive:
      - Single prompt (generate)
      - Multiple prompts for final evaluation (getp)
  - https://huggingface.co/openai/gpt-oss-20b
  - https://huggingface.co/openai/gpt-oss-120b

Project Settings

- **Baseline:** Provided CPU-only code (C/C++) — single-prompt, greedy decoding
- **Tokenizer:** Compatible with OpenAI o200k_harmony
- **Models:** Pre-exported binary weights from Hugging Face checkpoints

Project Goals

- Implement an inference serving system based on **llama2.c** for **gpt-oss** (20B & 120B)
  - Support **single-node** and **multi-GPU** execution
- Prioritize **correctness** above all else
  - Maintain metrics for correctness verification
- Achieve **highest throughput** (tokens/sec)
  - Apply diverse parallelization and optimization techniques

Technical Requirements

- C/C++ gpt-oss inference code provided
  - **Do not modify: run.cpp, run_eval.cpp, Makefile**
  - Add new C/C++ files via run_exec.cpp
- Parallelize & optimize with:
  - 4 nodes
  - Up to 8× AMD MI250 GPUs (HIP for GPU)
  - OpenMP / pthreads for CPU
- **Single-node only**
- Write all GPU kernels from scratch — no GPU libraries

Server Information (WIP)

- Login:
  - ssh getpXX@<server-ip>
  - Password: <passwd>
    - Change on first login (passwd)
- Job Execution (Slurm):
  - srun --gres=gpu:<N> <cmd>

Code & Resources (WIP)

- Code:
  - /mnt/getp/final-project/llama2.base_code.tar.gz
- Model Binaries:
  - /mnt/getp/final-project/modelbin
    - gpt-oss-7M.bin (debug only)
    - gpt-oss-20B.bin
    - gpt-oss-120B.bin
- Usage:
  - make # compile run.cpp → ./run
  - ./run model.bin -m generate -i “<input_prompt>”
  - ./run model.bin -m getp -f <input_file> -o <output_file>

References

- https://openai.com/index/introducing-gpt-oss/
- https://github.com/karpathy/llama2.c
- https://huggingface.co/openai/gpt-oss-20b
- https://github.com/openai/tiktoken
- https://rocmdocs.amd.com/en/latest/
- https://www.openmp.org/specifications/